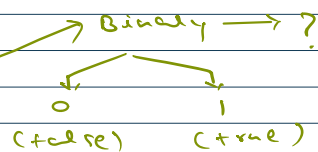
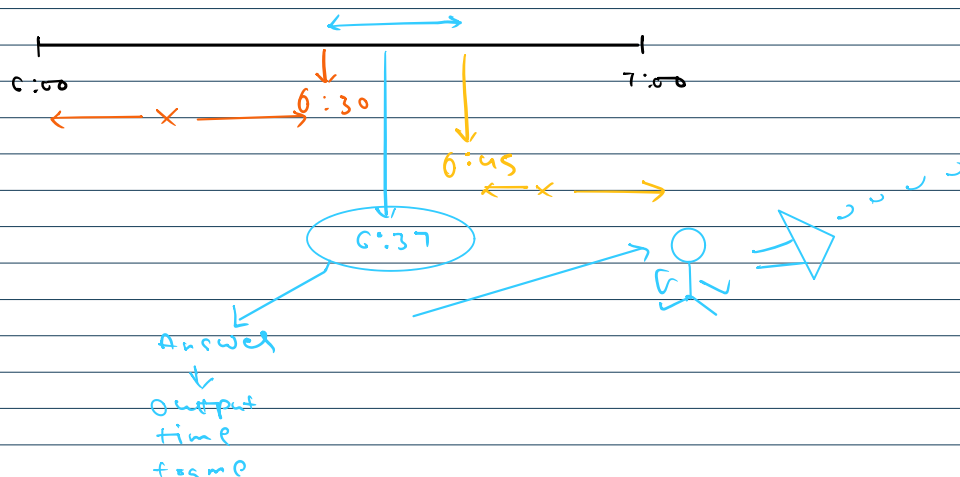


searching algorithms

Binary Search



Incident $\rightarrow$?



Binary search is an efficient algorithm for finding an element in a sorted list by repeatedly dividing the search range in half.

0	1	2	3	4	5	6	7	8	9	10	11	12	13
2	3	4	5	7	9	11	13	16	18	20	22	24	26

item: 13

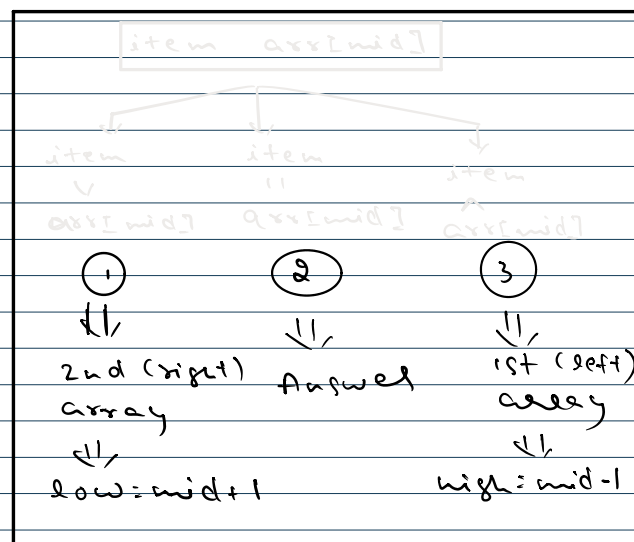
low $\rightarrow$ 0
high $\rightarrow$ 13 (4-1)

low = 0
high = 13 $\rightarrow$ mid = $\frac{low + high}{2} = \frac{0 + 13}{2} = 6$

13 > 11 $\Rightarrow$ ① $\Rightarrow$ 2nd array

low = 7 $\rightarrow$ mid + 1
high = 13

7	8	9	10	11	12	13
13	16	18	20	22	24	26

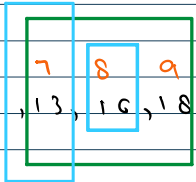


$$\begin{aligned} \text{low} &= 7 \\ \text{high} &= 13 \end{aligned} \rightarrow \text{mid} = \frac{(7+13)}{2} = 10$$

$$13 < 20 \Rightarrow \text{3rd}$$

$$\text{low} = 7$$

$$\text{high} = 9 \rightarrow \text{mid} = 1$$



$$\begin{aligned} \text{low} &= 7 \\ \text{high} &= 9 \end{aligned} \rightarrow \text{mid} = \frac{(7+9)}{2} = 8$$

$$13 < 16 \Rightarrow \text{3rd} \Rightarrow \text{left array (1st array)}$$

$$\begin{aligned} \text{high} &= \text{mid} - 1 \\ \text{high} &= 8 - 1 = 7 \end{aligned}$$



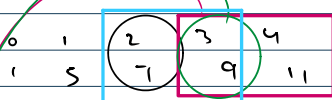
$$\begin{aligned} \text{low} &= 7 \\ \text{high} &= 7 \end{aligned} \rightarrow \text{mid} = \frac{(7+7)}{2} = 7$$

arr[mid], item

$$13 \neq 13 \Rightarrow \text{equal} \Rightarrow \text{2nd} \rightarrow \text{Answer} = \text{mid}$$

Loop exit condition $\Rightarrow \text{low} \leq \text{high} \Rightarrow \text{continue}$

Otherwise $\rightarrow \text{stop}$.



$$l \rightarrow 0$$

$$h \rightarrow 4$$

$$\text{item} = 8$$

$$\text{item} > \text{arr[mid]}$$

$$8 > 7 \Rightarrow \checkmark$$

$$m = \frac{(l+h)}{2} = 2$$

$$\begin{aligned} l &= 3 \\ h &= 4 \end{aligned} \rightarrow \text{mid} = \frac{(3+4)}{2} = 3$$

$$\text{item} < \text{arr[mid]} \Rightarrow 8 < 9 \Rightarrow \checkmark$$



$$\text{low} \leq \text{high} \Rightarrow \checkmark$$

$l = 1 \rightarrow mid = 4 \Rightarrow 43 < 149 \Rightarrow 3rd \Rightarrow l = 4 + 1 = 5$
 $h = 8$

$l = 5 \rightarrow mid = 6 \Rightarrow 63 > 149 \Rightarrow 1st \Rightarrow h = 6 - 1 = 5$
 $h = 8$

$l = 5 \rightarrow mid = 5 \Rightarrow 53 < 149 \Rightarrow 3rd \Rightarrow l = 5 + 1 = 6$
 $h = 5$

Answer
 //

Loop will stop

$l = 6$
 $h = 5$

~~low > high~~

Search in Sorted Rotated Array

arr $\rightarrow$ 0 1 2 4 5 6 7

item = 7

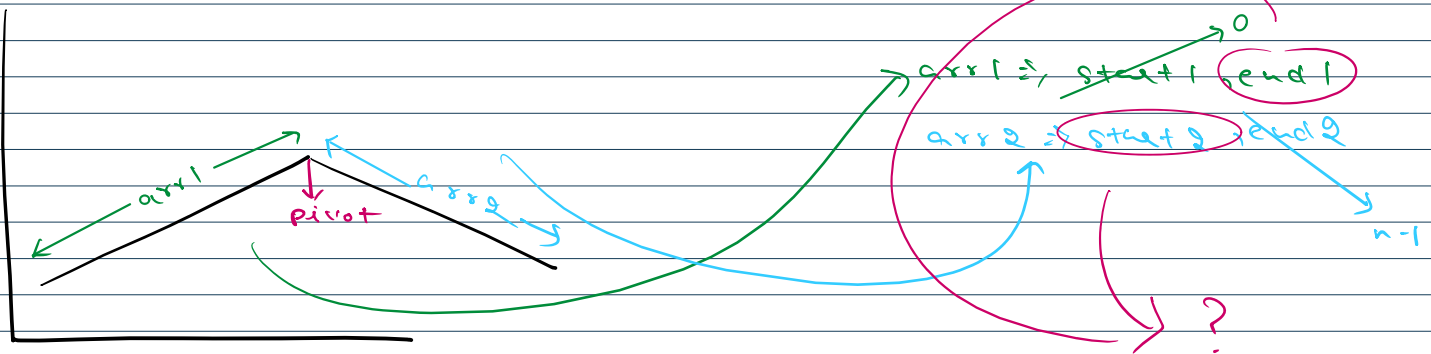
rotated arr $\rightarrow$ 4 5 6 7 0 1 2 $\Rightarrow$ Input

0	1	2	3	4	5	0
4	5	6	7	0	1	2

arr 1

arr 2

binary search
 item
 Ans
 binary search
 Ans



```

class Solution {
    public int search(int[] nums, int target) {
        int n = nums.length;

        int start1 = 0;
        int end1 = 0;
        int start2 = 0;
        int end2 = n-1;

        for(int i=0; i<n-1; i++) {
            if(nums[i] > nums[i+1]) {
                end1 = i;
                start2 = i+1;
            }
        }
    }
}

```

